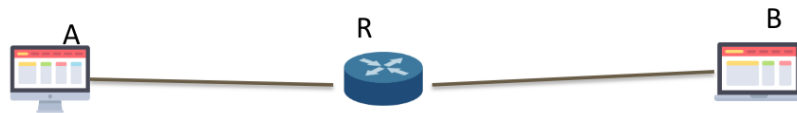


1 True or False

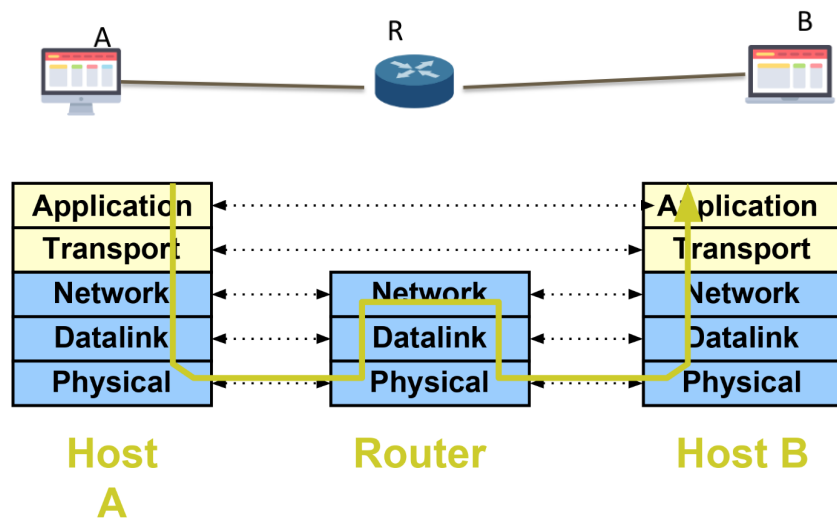
- (1) The third layer of the network stack is the transport layer.
- (2) Layering in the network stack is an example of the end-to-end principle.
- (3) We implement the same layers in both the host and routers.
- (4) The operating system (OS) supports logical ports while a router only has physical ports

2 Network Stack

Consider host A that sends a packet to host B in the following diagram. In this problem will track what happens to the headers of the packet as it moves through the network.



- (1) What is the network stack?
- (2) Does a switch need layers 4 and 5? Why or why not?
- (3) Host A's packet is now ready to be sent across a link. Which headers does the packet have and what is the order?
- (4) When Host A's packet arrives at a router as bits, how do the headers of the packet change?

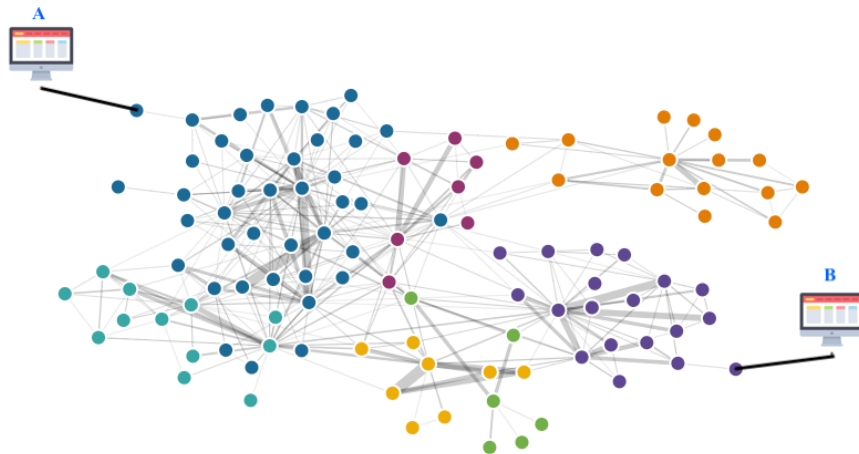


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3 Designing the Data Plane

Consider designing the internet from first principles. We'll go over how to create a design by considering how to solve the problem that lead to the modern day internet.

For this problem, assume we have many hosts communicating with each other over the network.



- (1) What problem the internet is trying to solve?
- (2) What are the fundamental constraints hosts face when communicating data over a network whose resources are shared with other hosts? Create a list of the fundamental constraints of the internet. (We'll be going over this list of constraints as we progress through the semester.)

- (3) In the modern day internet, what are the physical components in a network where two hosts are communicating?
- (4) Focusing on routing on the data plane, how would you break up communication into smaller tasks (Note this is for a best effort design for the internet)?
- (5) Recall the network stack of the internet. Where in the network stack should reliability, routing, and security be implemented? Consider designing functionality in only a single layer and where the state of each flow should be shared. (Note: these considerations lead to significant design decisions in the modern day internet.)

